Lab Exercise 19- ClusterIP in Service in Kubernetes

A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels)

Create file:

nginx-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx   # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx  # MUST match service selector
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Save file.

```
kubectl apply -f nginx-deployment.yaml
```

Verify:

```
kubectl get pods --show-labels
```

You should see:

```
app=nginx
```

nginx-deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx   # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx  # MUST match service selector
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

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```
PowerShell 7.5.4
PS C:\Windows> cd C:\Users\HP\Desktop\COAS_Lab\Lab_8
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-app created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get pods --show-labels
NAME                          READY   STATUS    RESTARTS      AGE   LABELS
nginx-app-86c57bc6b8-s97p4    1/1     Running   0             9s    app=nginx,pod-template-hash=86c57bc6b8
nginx-app-86c57bc6b8-z2zxq    1/1     Running   0             9s    app=nginx,pod-template-hash=86c57bc6b8
test-vol1                     1/1     Running   1 (27h ago)   27h   <none>
test-vol2                     1/1     Running   1 (27h ago)   27h   <none>
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx   # MUST match pod label
  ports:
  - port: 80
    targetPort: 80
```

Apply:

```
kubectl apply -f nginx-service.yaml
```

```yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx   # MUST match pod label
  ports:
  - port: 80
    targetPort: 80
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl apply -f nginx-service.yaml
service/nginx-service created
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 3: Verify It Is Working

Check Service

```
kubectl get svc
```

You should see:

```
TYPE        CLUSTER-IP
ClusterIP   10.x.x.x
```

MOST IMPORTANT CHECK

```
kubectl get endpoints nginx-service
```

Now you should see something like:

```
10.244.x.x:80, 10.244.x.x:80
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get svc
NAME            TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
kubernetes      ClusterIP   10.96.0.1       <none>        443/TCP   21d
nginx-service   ClusterIP   10.96.238.152   <none>        80/TCP    29s
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl get endpoints nginx-service
NAME            ENDPOINTS                     AGE
nginx-service   10.1.0.124:80,10.1.0.125:80   50s
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 4: Test From Inside Cluster

Run test pod:

```
kubectl run test-pod --image=busybox --rm -it -- sh
```

Inside pod:

```
wget -qO- http://nginx-service
```

You should see:

```
Welcome to nginx!
```

Exit:

```
exit
```

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> kubectl run test-pod --image=busybox --rm -it -- sh
All commands and output from this session will be recorded in container logs, including credentials and sensitive information passed through the command pro
mpt.
If you don't see a command prompt, try pressing enter.
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit
Session ended, resume using 'kubectl attach test-pod -c test-pod -i -t' command when the pod is running
pod "test-pod" deleted from default namespace
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

```
curl http://<CLUSTER-IP>
```

It will NOT work.

That proves:

ClusterIP works only inside cluster.

```
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8> curl http://10.96.238.152
curl: (28) Failed to connect to 10.96.238.152 port 80 after 21048 ms: Could not connect to server
PS C:\Users\HP\Desktop\COAS_Lab\Lab_8>
```